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|---------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------|------|
| Test Case Project                     | Food Waste Reducer Application - Team 47                               | Test Case Description                                          | Exhaustive verification of full-stack functionalities for food waste reducer website. |                                        |      |
| Created By                            | Chalavadi Gayathri - SE23UCSE047                                       | Reviewed By                                                    | G Siddanth Reddy - SE23UCSE061                                                        | Version                                | 1    |
| Tester's Name                         | Chalavadi Gayathri                                                     | Date Tested                                                    | April 28, 2026                                                                        | Overall Test Case Status               | Pass |
|                                       |                                                                        |                                                                |                                                                                       |                                        |      |
| QA Tester's log                       | Final review completed. All FCM and Cron-job functionalities verified. |                                                                |                                                                                       |                                        |      |
|                                       |                                                                        |                                                                |                                                                                       |                                        |      |
| Test Scenario                         |                                                                        |                                                                |                                                                                       |                                        |      |
|                                       |                                                                        |                                                                |                                                                                       |                                        |      |
| Step #                                | Step Details                                                           | Expected Results                                               | Actual Results                                                                        | Pass / Fail / Not executed / Suspended |      |
| 1. Authentication & Security          |                                                                        |                                                                |                                                                                       |                                        |      |
| 1                                     | Add new user via registration.                                         | Opens a form with fields: Name Role Password.                  | As expected                                                                           | Pass                                   |      |
| 2                                     | Login with JWT generation.                                             | Successful login and token stored.                             | As expected                                                                           | Pass                                   |      |
| 3                                     | Verify password hashing.                                               | Passwords stored as hashed strings via Bcrypt.                 | Verified in MongoDB                                                                   | Pass                                   |      |
| 4                                     | Test protected routes.                                                 | Accessing inventory without login redirects to login.          | Unauthorized access blocked                                                           | Pass                                   |      |
| 2. Dashboard Module                   |                                                                        |                                                                |                                                                                       |                                        |      |
| 5                                     | Verify dashboard summary cards populate.                               | Display summary cards and "Expiring Soon" preview.             | Verified dashboard summary cards populate.                                            | Pass                                   |      |
| 6                                     | Check dashboard for real-time alerts.                                  | Notifications preview section shows latest alerts.             | Alerts visible                                                                        | Pass                                   |      |
| 3. Inventory Module                   |                                                                        |                                                                |                                                                                       |                                        |      |
| 7                                     | Add food item with expiry date.                                        | Item stored in Database and displayed in list.                 | As expected                                                                           | Pass                                   |      |
| 8                                     | Retrieve User-Specific Inventory                                       | Only items in certain user's inventory are shown from DB.      | Filters working correctly                                                             | Pass                                   |      |
| 9                                     | Verify color-coded expiry logic.                                       | Items expiring soon show specific tags/icons.                  | As expected                                                                           | Pass                                   |      |
| 10                                    | Remove an item from the list.                                          | Item removed from UI and MongoDB instantly.                    | Sync successful                                                                       | Pass                                   |      |
| 4. Meal Planner Module                |                                                                        |                                                                |                                                                                       |                                        |      |
| 11                                    | Dynamic Recipes shown on user dashboard                                | Meal recipes shown on UI according to user inventory.          | As expected                                                                           | Pass                                   |      |
| 12                                    | Generate recipe suggestions from expiring food.                        | Business logic suggests meals using current stock.             | As expected                                                                           | Pass                                   |      |
| 5. Notifications Module               |                                                                        |                                                                |                                                                                       |                                        |      |
| 13                                    | Trigger browser notification permission prompt.                        | Navigate to Alerts page and trigger prompt.                    | As expected                                                                           | Pass                                   |      |
| 14                                    | Generate and save FCM token in MongoDB.                                | Unique FCM token saved to User DB on 'Allow'.                  | As expected                                                                           | Pass                                   |      |
| 15                                    | Verify foreground alert pop-up.                                        | Alert pop-up appears while using the app.                      | As expected                                                                           | Pass                                   |      |
| 16                                    | Confirm background system-level notification (Plant Icon).             | Desktop banner shown when tab is closed.                       | As expected                                                                           | Pass                                   |      |
| 17                                    | Run 09:00 AM Cron job automation test.                                 | Server triggers expiry scan and sends alerts.                  | As expected                                                                           | Pass                                   |      |
| 6. Storage & Sharing Module           |                                                                        |                                                                |                                                                                       |                                        |      |
| 18                                    | View food storage tips interface.                                      | Display researched tips on extending shelf life.               | As expected                                                                           | Pass                                   |      |
| 19                                    | Look for food composting/donation options.                             | Info. on food sharing programs & further interaction via mail. | As expected                                                                           | Pass                                   |      |
| 20                                    | Post an Item for Sharing                                               | User can list excess food for community pickup                 | Item posted to sharing feed                                                           | Pass                                   |      |
| 21                                    | Request an Item                                                        | User can view and request items posted by others               | Request successfully logged                                                           | Pass                                   |      |
| 7. Responsive UI & API Error handling |                                                                        |                                                                |                                                                                       |                                        |      |
| 22                                    | Trigger and handle API errors.                                         | Application handles API errors with user messages.             | Error messages shown                                                                  | Pass                                   |      |
| 23                                    | Responsive UI Layout                                                   | Navigation and assets adapt to mobile/desktop                  | Verified across devices                                                               | Pass                                   |      |